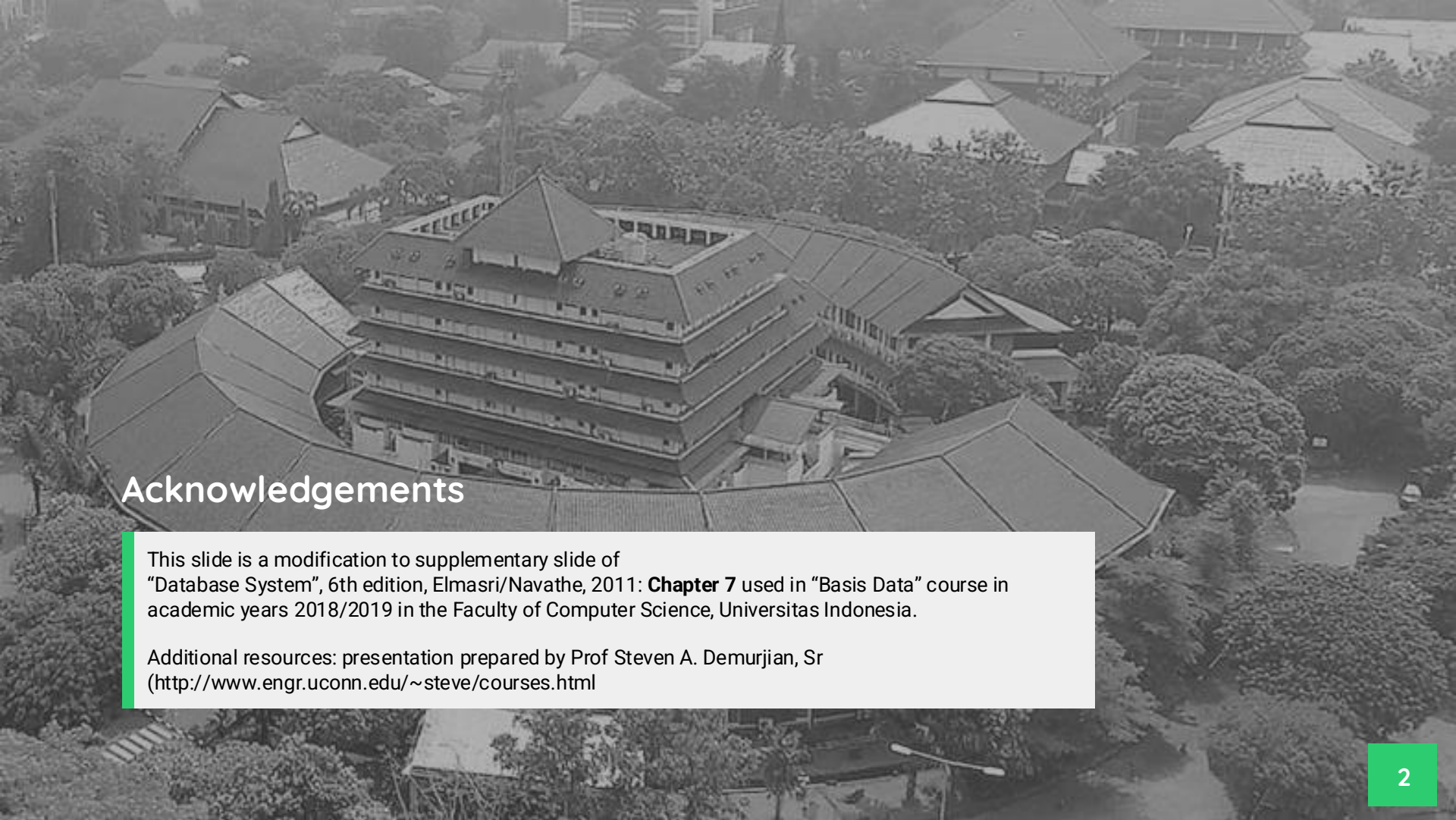
3

Pemodelan Basis Data dengan ER

(PART 2)

CSGE602070 - BASIS DATA



Acknowledgements

This slide is a modification to supplementary slide of
“Database System”, 6th edition, Elmasri/Navathe, 2011: **Chapter 7** used in “Basis Data” course in academic years 2018/2019 in the Faculty of Computer Science, Universitas Indonesia.

Additional resources: presentation prepared by Prof Steven A. Demurjian, Sr
(http://www.engr.uconn.edu/~steve/courses.html

Yang sudah dipelajari

Yang sudah dipelajari:

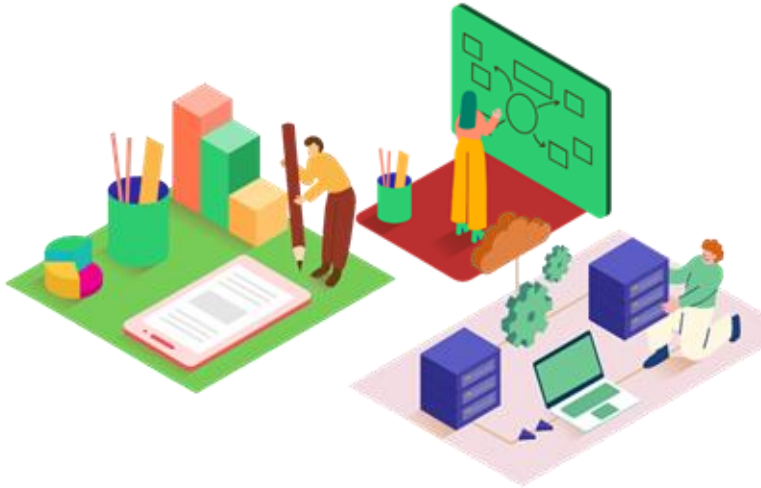
Tahapan pengembangan Basis Data

Contoh aplikasi Basis Data: Studi kasus basis data Company

Konsep model ER: entity, atribute, key atributte

Outline

1. Tahapan Pengembangan Basis Data
2. Contoh Aplikasi Basis Data
3. Konsep-konsep model ER
4. Konsep model ER: relationship, weak entity
5. Pembuatan diagram ER



Memetakan *Requirement* ke Awal Konsep *Entity* untuk Basis Data COMPANY

Data pribadi **PEGAWAI** meliputi SSN, name (first name, middle name, last name), address, salary, sex, birthdate

- ➔ Tiap **EMPLOYEE bekerja pada satu DEPARTMENT**, namun dapat terlibat pada beberapa PROJECT, yang tidak harus di department yang sama dengan pegawai.
- ➔ Jumlah jam kerja pegawai pada tiap project dicatat
- ➔ Supervisor langsung dari setiap EMPLOYEE juga dicatat

EMPLOYEE
Name (EName, MInit, LName), SSN, Sex, Address, Salary, BirthDate, Department, Supervisor, {Workson(Project, Hours)}

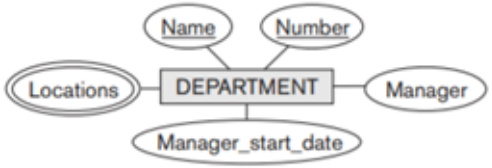
Pegawai dapat memiliki tanggungan (**DEPENDENTs**)

- ➔ Dicatat nama, jenis kelamin, tanggal lahir, dan hubungan tanggungan dengan pegawai

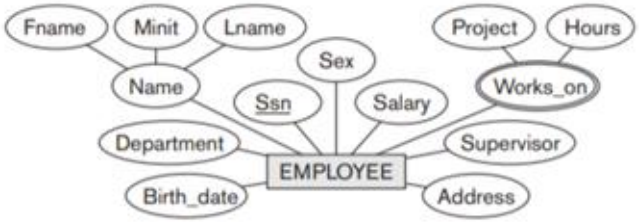
DEPENDENT
Employee, DependentName, Sex, Birthdate, Relationship

Awal Konsep Entity untuk Basis Data COMPANY

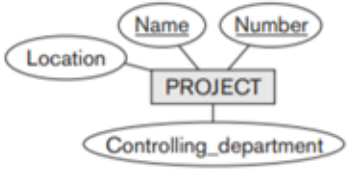
DEPARTMENT
Name, Number, Locations, Manager, ManagerStartDate



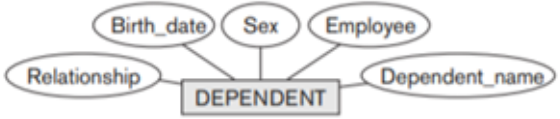
EMPLOYEE
Name (EName, MInit, LName), SSN, Sex, Address, Salary, BirthDate, Department, Supervisor, {Workson (Project, Hours)}



PROJECT
Name, Number, Location, ControllingDepartment



DEPENDENT
Employee, DependentName, Sex, Birthdate, Relationship



Relationship

Sebuah relationship menghubungkan dua/lebih entity yang berbeda dengan **makna** tertentu.

Contoh:

- ➔ EMPLOYEE John Smith ***works on*** the ProductX PROJECT
- ➔ EMPLOYEE Franklin Wong ***manages*** the Research DEPARTMENT.

Relationships yang bertipe sama dikelompokkan ke dalam sebuah **relationship type**.

Contoh:

- ➔ **WORKS_ON** relationship type in which EMPLOYEEs and PROJECTs participate
- ➔ **MANAGES** relationship type in which EMPLOYEEs and DEPARTMENTs participate

Relationships dan Relationship Types

Derajat dari Relationship Type adalah jumlah Entity Types yang berpartisipasi pada relationship ini (**Binary**, **Ternary**, **N-nary**)

Satu entity type dapat berpartisipasi pada lebih dari satu relationship type Relationships memiliki hubungan directional (arah tertentu):

- ➔ SUPPLIES: SUPPLIER to PARTS
- ➔ SUPPLIERS: PARTS to SUPPLIER

Binary Relationship

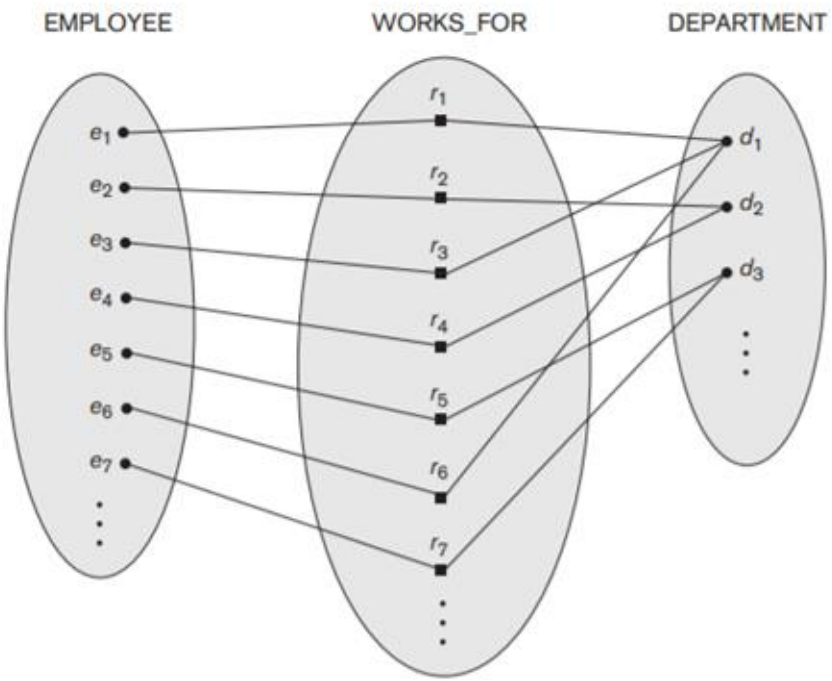


Figure 3.9
Some instances in the WORKS_FOR relationship set, which represents a relationship type WORKS_FOR between EMPLOYEE and DEPARTMENT.

Ternary Relationship

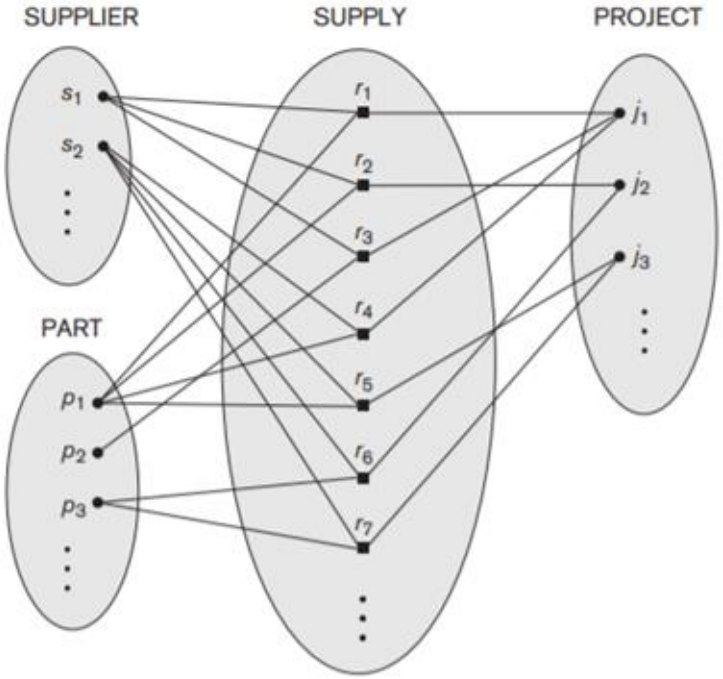


Figure 3.10
Some relationship instances in the SUPPLY ternary relationship set.

Structural Constraints

Digunakan untuk mengekspresikan semantik dari relationship.
Ada 2 macam:

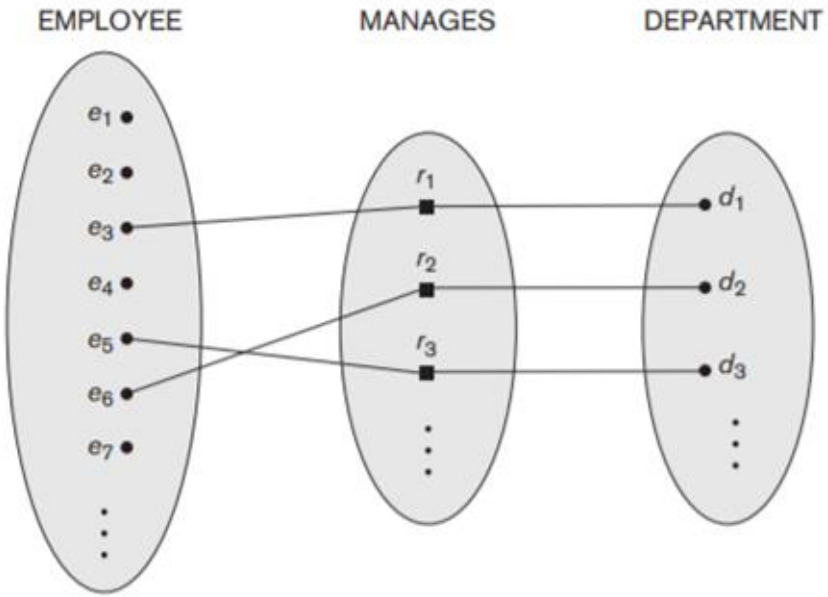
- **Cardinality ratio** (dari binary relationship): 1:1, 1:N, N:1, M:N
  Ditunjukkan dengan memberikan **nomor** yang sesuai pada link dari relationship.

- **Participation constraint**: Total, partial

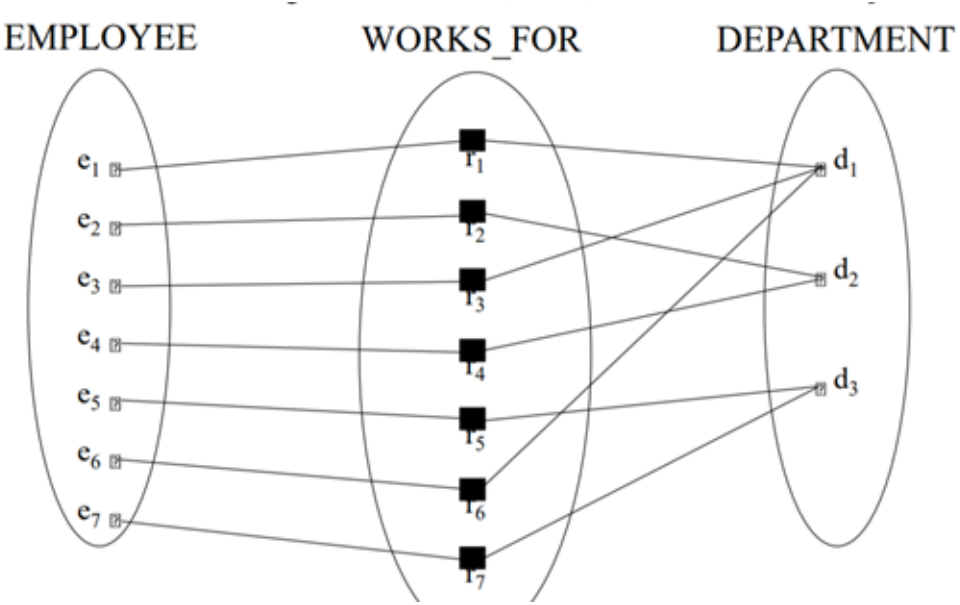
  - Double link untuk **total** participation: minimal ada 1 entity set yang terlibat pada relationship.
  - Single link untuk **partial** participation: diperbolehkan apabila tidak ada entity set yang terlibat pada relationship

One-to-One (1:1) Relationship

Figure 3.12
A 1:1 relationship, MANAGES.



Many-to-One (N:1) Relationship



Many-to-Many (M:N) Relationship

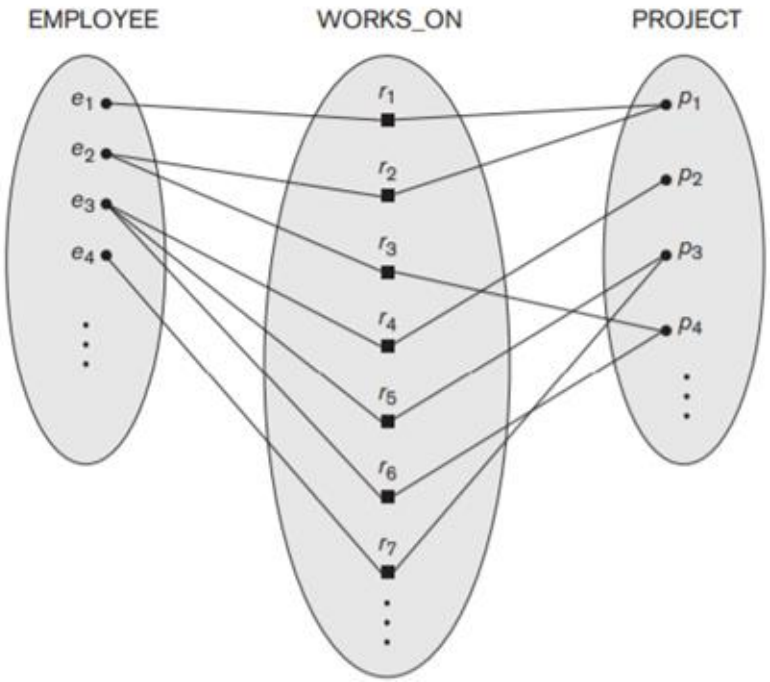
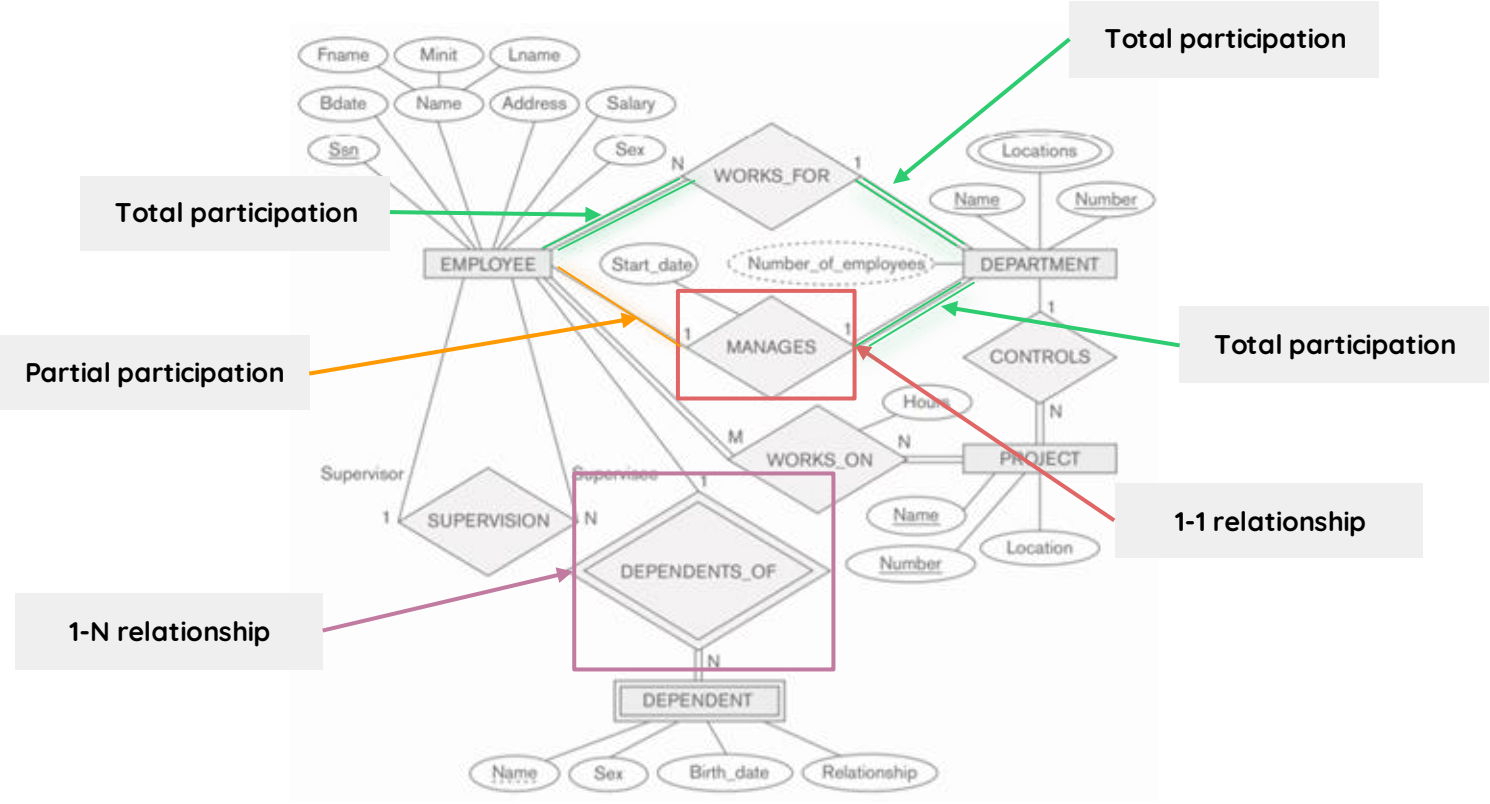


Figure 3.13
An M:N relationship, WORKS_ON.

Structural Constraints



Recursive Relationship

Satu entity **berpartisipasi lebih dari 1 kali** dalam recursive relationship, dengan **role yang berbeda**

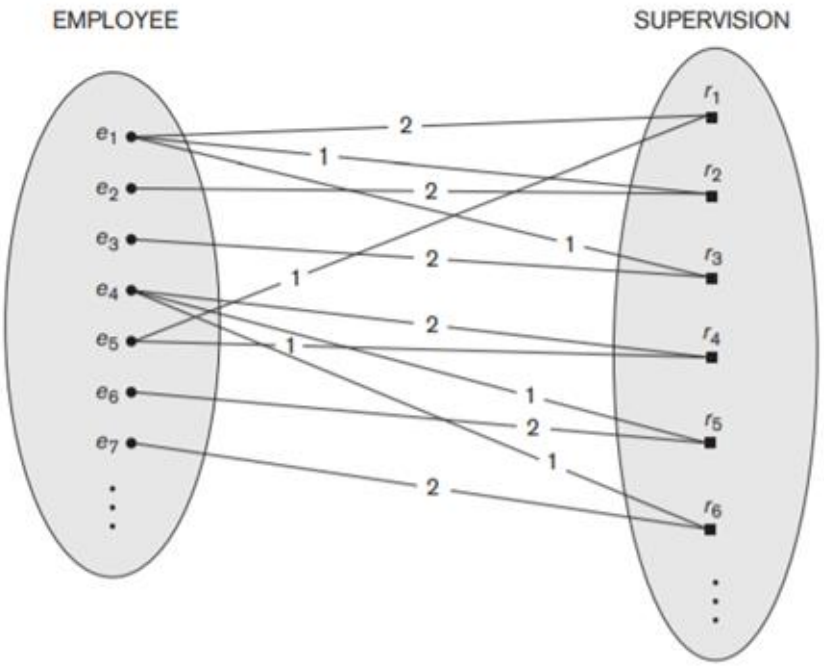
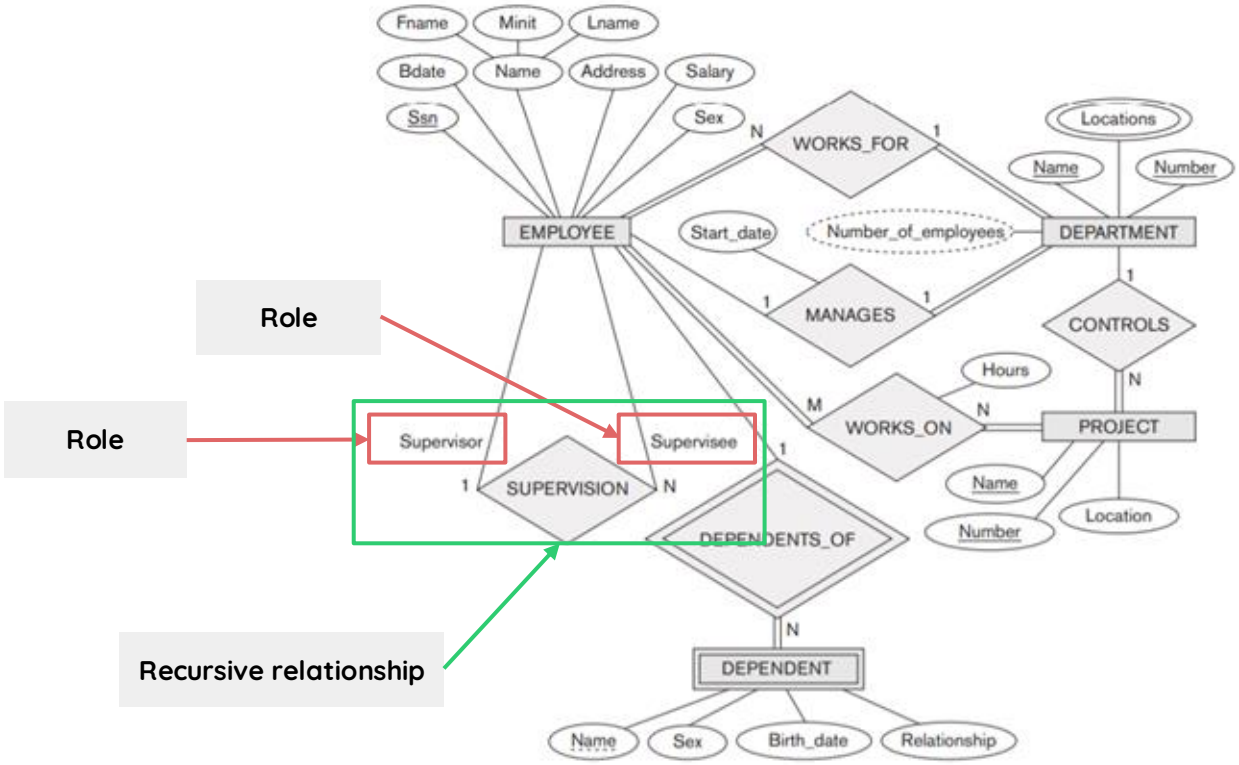
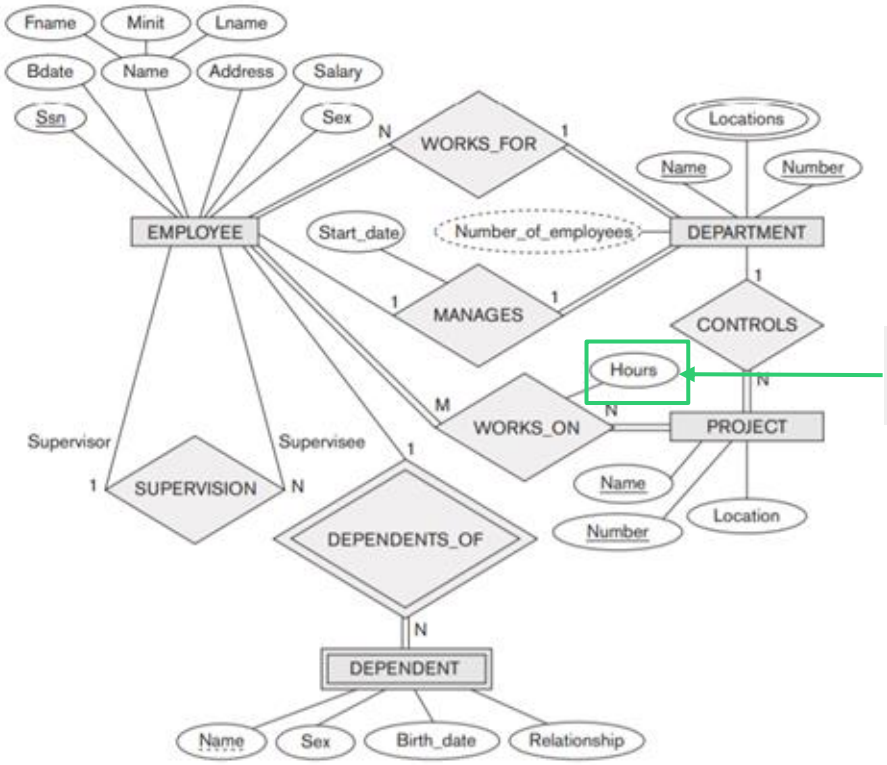


Figure 3.11
A recursive relationship SUPERVISION between EMPLOYEE in the *supervisor* role (1) and EMPLOYEE in the *subordinate* role (2).

Contoh Recursive Relationship



Attribute dari Relationship Type



Attribute untuk WORKS_ON

Menentukan Relationship

Uji kombinasi dari 2 entity, lihat **apakah ada kemungkinan relationship** diantara keduanya.

Lihat dokumen **requirements**

EMPLOYEE	?	DEPENDENT
DEPARTMENT	?	PROJECT
EMPLOYEE	?	DEPARTEMENT

Dan lainnya ...

Weak Entity Types

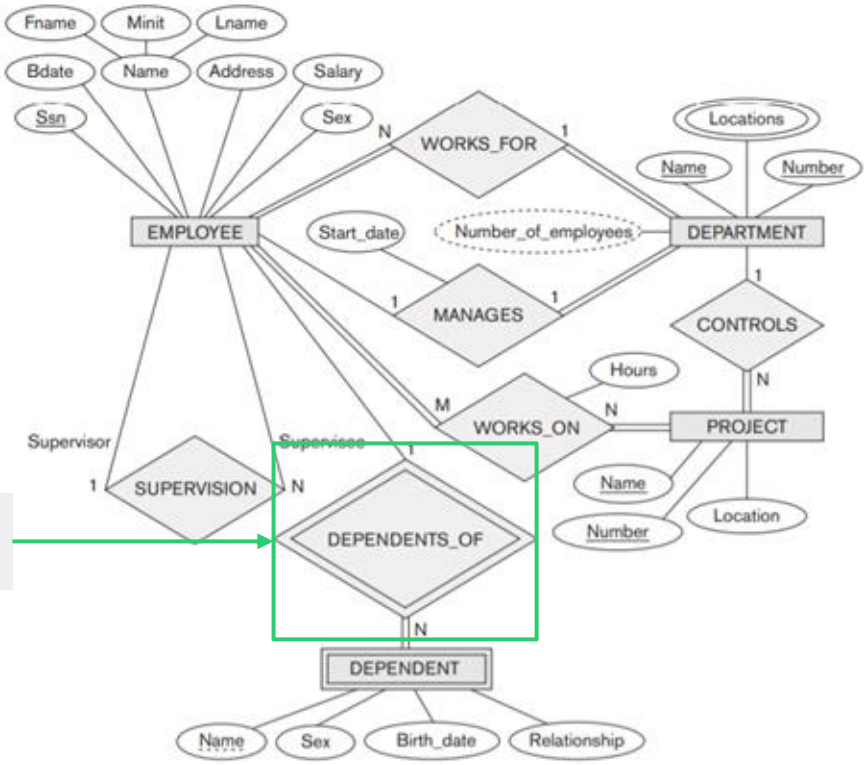
Merupakan entity yang **tidak punya key attribute**

Weak entity **harus berpartisipasi** dalam sebuah identifying relationship type dengan suatu owner atau identifying entity type

Weak entity diidentifikasi dengan kombinasi dari:

- ➔ Partial key dari weak entity type
- ➔ Attribute pada owner entity yang menjadi penghubung antara weak entity dengan owner-nya

Contoh Weak Entity Types



Attribute untuk WORKS_ON

Alternatif Notasi (min, max) untuk Relationship Structural Constraints

Dinyatakan pada setiap partisipasi dari sebuah *entity type* E pada *relationship type* R

Menyatakan bahwa tiap entity e di E berpartisipasi paling sedikit min dan paling banyak **max** pada relationship instance di R

Menyatakan batasan yang ada dalam requirement

Harus memenuhi

- ➔ min max, min 0, max 1

Default (tidak ada batasan)

- ➔ min = 0
- ➔ max = n

min = 0 menandakan partisipasi parsial
min 1 menandakan partisipasi total

Alternatif Notasi (min, max) untuk Relationship Structural Constraints

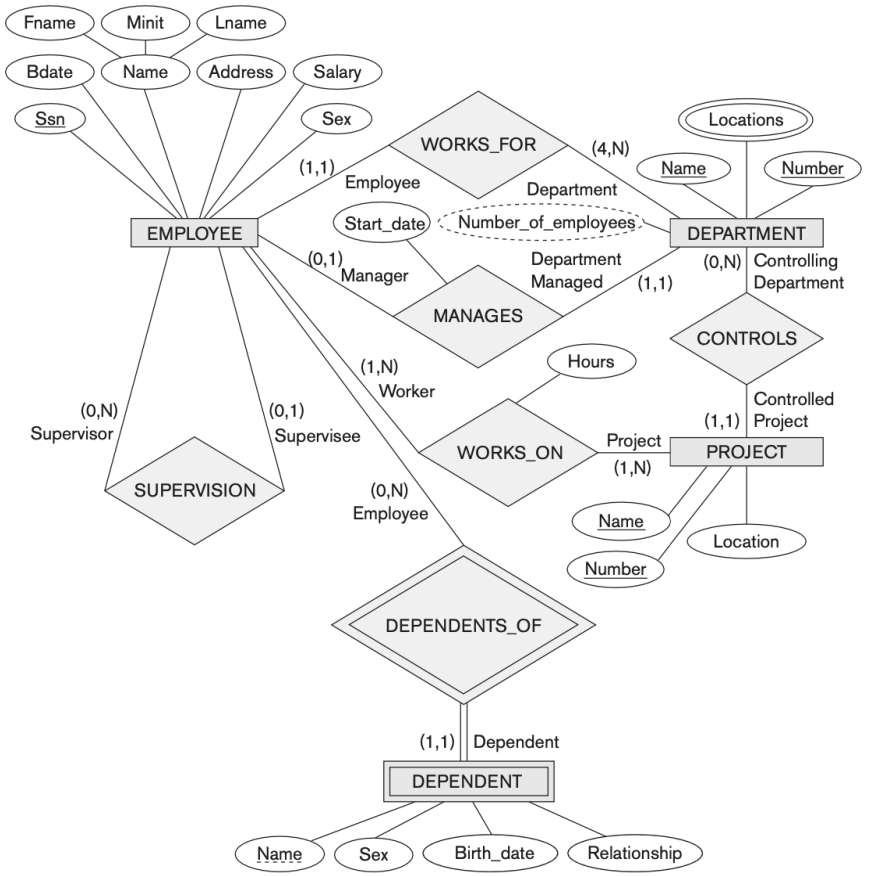


Figure 3.15
ER diagrams for the company schema, with structural constraints specified using (min, max) notation and role names.

Q&A